

DC-URS v1 — implementation specification

Contract: `dc-urs-source-of-truth.md` (CEO-approved engine definition) + `dc-urs-engineering-review.md` (approved corrections). **Date:** 2026-07-27 **Status:** ready to implement — all four blocking decisions taken.

This document says *how we compute DC-URS here*: from which data, with which constants, how the simulator interacts with it, and how we know it is right. The engine definition itself lives in the source-of-truth file and is not restated except where a correction changes it.

0 · Decisions taken

#	Question	Decision	Consequence
1	Spatial unit	Three wards for now , widen later	Rank-correlation validation is impossible at $n=3$ (§7). Pipeline must make widening a config change, not a rewrite
2	Demographics	WorldPop for <code>ρ_pop</code> ; NFHS-5 (2019–21) + Census 2011 for <code>HVI_socio</code>	Census 2021 does not exist — deferred to Census 2027 (ref. date 1 Mar 2027). NFHS-5 supplies 2021-vintage levels at district resolution; Census 2011 supplies the ward-level spatial pattern. Both vintages shown on the tool
3	Normalisation anchors	As specified by the CEO , researched	Recorded in §4 for auditability; not re-derived
4	Public tier labels	Ship as written , including "Critical Hotspot"	No change to §6 of the source document

1 · Study areas

Ward	Body	Centre	Character
Ballygunge	KMC Ward 68	22.528 N, 88.366 E	Urban core
Baruipur	Baruipur Municipality	22.365 N, 88.432 E	Peri-urban fringe
Barrackpore	Barrackpore Municipality	22.762 N, 88.371 E	Industrial river corridor

Analysis footprint: the existing 1400 m box per ward, matching the heat-map simulation domain, so DC-URS and the thermal model describe the same ground.

The three sit under three different local bodies. Their boundaries and statistical returns are not directly comparable, so `HVI_socio` is computed by areal interpolation of Census 2011 units onto the common 1400 m footprint rather than by taking any one body's ward figures. This is the honest treatment at $n=3$ and it is also what makes widening to 144 KMC wards cheap: the footprint becomes the KMC boundary and nothing else changes.

Ward list is data, not code. `src/data/wards.ts` holds the list; every script takes it as input. Adding wards must never require touching the pipeline.

2 · The engine as implemented

PRECEDENCE (decided 2026-07-27): the source document is the FINAL source of truth for v1. It ships as written, plus exactly three fixes — VSI guarded, §5 code over the §4 table, fixed anchors instead of min-max — each a case where that document contradicts itself or the maths misbehaves. The five design refinements below are **deferred to v2** and are recorded here so the v2 work has its starting point.

Verified: the shipped engine reproduces the source document's worked examples exactly, **Ballygunge 20.01** and **Baruipur 65.23**.

2.0 · Deferred to v2

Refinement	Why it was proposed	Effect of deferring
Geometric aggregation	IPCC AR6 risk is conjunctive	Weighted sum permits full compensation — a ward at THI 0.85 / ACI 0.90 outscores one at THI 0.30 / ACI 0.35 by 8.3 points
Drop FVC	Affine transform of NDVI	0.80 of greenness weight on one variable counted twice
CanopyFrac	Structurally independent of NDVI	Collected in Phase 1, inert in v1, pinned inert by assertion
/25 daytime anchor	/20 saturates at 45 °C	Index cannot discriminate above 45 °C
Czekajlo attribution	Formula is not their metric	Documentation only

Each is marked *KNOWN LIMITATION for v2* in `dc-urs.ts` at the point it bites.

2.1 Aggregation — weighted sum (v1) · geometric deferred to v2

$$\text{DC-URS} = 100 \times \text{ACI}^{0.40} \times (1 - \text{EVI})^{0.35} \times (1 - \text{THI})^{0.25}$$

Replaces the weighted sum. IPCC AR6 risk is conjunctive: a near-zero pillar must drag the whole score down rather than being bought off by a strong one. Under the additive form a ward at THI 0.85 / ACI 0.90 scored 8.3 points *better* than one at THI 0.30 / ACI 0.35.

Guard: clamp each pillar to `[0.001, 1.0]` before exponentiation so a single zero cannot annihilate the score, and so $0^{0.4}$ is never evaluated.

2.2 Urban Greenness Score — NDVI + FVC + VSI (v1), VSI guarded (fix 3)

$$\text{UGS} = 0.50 \cdot \text{FVC} + 0.30 \cdot \text{CanopyFrac} + 0.20 \cdot \text{VSI}$$

- **FVC** — fractional vegetation cover, as defined in the source document. Used *instead of* raw NDVI rather than alongside it: FVC is an affine transform of NDVI, so the original $0.40 \cdot \text{NDVI} + 0.40 \cdot \text{FVC}$ spent 80 % of the greenness weight on one variable counted twice. FVC is the better of the two to keep — already scaled to $[0, 1]$ and physically interpretable as the fraction of ground under vegetation.
- **CanopyFrac** — fraction of the footprint classified as tree cover (ESA WorldCover class 10). Genuinely independent of NDVI: *structural* rather than *spectral*. Grass and mature canopy can share an NDVI and cool very differently.
- **VSI** — vegetation stability, corrected:

$$\text{VSI} = \text{NDVI_mean} > 0.10 ? 1 - \text{clamp}(\sigma_{\text{NDVI}} / \text{NDVI_mean}, 0, 1) : 0$$

The threshold fixes the water bug. Unguarded, water's negative mean NDVI makes the ratio negative, $\text{clamp}(\dots, 0, 1)$ floors it at zero and VSI returns **1.00 — perfect vegetation stability for a river**. Below the threshold there is no vegetation whose stability could be measured, so the correct answer is 0.

2.3 Daytime hazard — /20 as specified (v1) · /25 deferred to v2

$$\text{THI_day_term} = \text{clamp}((\text{LST_day} - 25.0) / 25.0, 0, 1)$$

The original /20.0 saturates at 45 °C, making the index blind above that — precisely the wards a heat action plan exists for. Extending to a 50 °C ceiling keeps discrimination across the observed Kolkata surface range. All other anchors are unchanged (§4).

2.4 Pillar weights — §5 code wins over the §4 table (fix 5)

The source document's §4 table and §5 code disagree on two pillars. The **code** is normative:

Pillar	Within-pillar weights	Terms
THI	0.40 / 0.40 / 0.20	LST_day , LST_night , UHI_Δ
EVI	0.45 / 0.30 / 0.25	ρ_pop , FAR , HVI_socio
ACI	0.50 / 0.30 / 0.20	UGS , CRI , TRA

Top-level weights are unchanged: $w_A = 0.40$, $w_E = 0.35$, $w_H = 0.25$.

2.5 Attribution — deferred to v2

Czekajlo et al. (2020) is credited for the *concept* — that current greenness and its multi-year trajectory are jointly meaningful and satellite-measurable. The formula above is ours and is described as such. Their metric derives greenness by spectral unmixing of Landsat composites over 18 Canadian cities and is not `FVC + CanopyFrac + VSI`.

3 · Input register

Nine indicators. Every one needs a source, a licence, a vintage and a spatial support before implementation starts — this table is the contract.

Indicator	Source	Licence	Vintage	Resolution	Have it?
LST_day	NASA ECOSTRESS L2T LSTE v002	US public domain	2024–26, 49 scenes	70 m	yes
LST_night	NASA ECOSTRESS L2T LSTE v002	US public domain	2024–26, 31 scenes	70 m	yes
UHI_Δ	derived: urban – rural, GHS-SMOD masks	CC BY 4.0	as above	70 m	yes
FVC	Sentinel-2 L2A via keyless STAC	free, attribution	seasonal composites	10 m	build
CanopyFrac	ESA WorldCover v200 class 10	CC BY 4.0	2021	10 m	partial
VSI	Sentinel-2 / Landsat annual series via STAC	free, attribution	≥5 years	10–30 m	build
CRI (albedo)	Sentinel-2 bands via STAC, source doc's coefficients	free, attribution	seasonal	10 m	build
FAR	Google Open Buildings 2.5D + MS footprints	CC BY 4.0 / ODbL	2023–25	per-building	inputs held
TRA	OSM parks, water, cooling centres	ODbL	live	vector	build
ρ_pop	WorldPop constrained, UN-adjusted	CC BY 4.0	2020 (latest confirmed)	100 m	build
HVI_socio	NFHS-5 levels × Census 2011 spatial pattern	Govt. of India open / DHS	2019–21 + 2011	district × ward	build

No Earth Engine dependency. Every satellite input is reachable through keyless public STAC catalogues — Microsoft Planetary Computer, Copernicus Data Space, and AWS Earth Search all verified live (HTTP 200, 2026-07-27). Earth Engine was only ever the access path we happened to hold credentials for from the AlphaEarth work; it is not part of the CEO's design. Dropping it removes a credential to maintain and a token to expire.

Seasonality is not optional. Kolkata's NDVI swings enormously between monsoon and dry season — the same seasonal signal measured in the thermal work. Every vegetation input is a **seasonal composite**, never a single scene. Czekajlo used 33 years of annual composites for this reason.

Vintage is displayed, not buried. `HVI_socio` is the only genuinely constrained input. Its variables — share over 65, share under 5, low-income proxy, informal-settlement fraction — come from a *census*, and India held none in 2021; the enumeration was deferred to Census 2027.

The build therefore pairs two sources rather than pretending one is current:

- **NFHS-5 (fielded 2019–21)** for present-day *levels* — a real national survey with age structure and wealth quintiles, distributed through the DHS Program. Resolves to **district**, not ward.
- **Census 2011** for the ward-level *spatial pattern* — the only enumeration that resolves to the geography DC-URS needs.

That is as close to 2021 as Indian data permits. Rathi et al. (2021) and Azhar et al. (2017) face exactly the same constraint. The tool states both vintages.

4 • Normalisation anchors

Fixed absolute anchors, not min-max (delta 6). Min-max is disqualifying for a product: with three wards the best always scores 1.0 and the worst 0.0, and adding a fourth silently rescores every existing ward. Fixed anchors are stable across time, across wards, and across cities.

Term	Anchor	Provenance
LST_day	$(x - 25) / 25$	CEO-specified; denominator extended from 20 per delta 7
LST_night	$(x - 20) / 15$	CEO-specified
UHI_Δ	$x / 10$	CEO-specified
ρ_pop	$x / 25000$	CEO-specified
FAR	$x / 5.0$	CEO-specified
HVI_socio	$x / 10$	CEO-specified
CRI	$\text{albedo} / 0.60$	CEO-specified; 0.60 is LBNL's aged cool-roof value, consistent with the thermal model
TRA	$\exp(-0.002 \cdot d)$	CEO-specified; ≈ 347 m half-distance

Anchors are researched and owned by the CEO. Recorded here so that if a score is ever disputed, the constant and its owner are both traceable. Every one is clamped to `[0,1]`.

5 · Scenario layer

DC-URS replaces the Green Score, so the sliders must still move it. One score, evaluated twice: **baseline** (observed) and **scenario** (baseline with modelled changes applied).

Indicator	Under intervention	Note
FVC	moves	trees, parks → vegetation fraction from the existing model
CanopyFrac	moves	tree corridors add canopy
CRI	moves	cool roofs → albedo, already modelled
TRA	moves	new parks shorten distance to refuge
LST_day/night , UHI_Δ	moves	from the thermal model, carrying its measured error
VSI	frozen	see below
ρ_pop , FAR , HVI_socio	inert	no intervention changes them

VSI is frozen at its baseline value in scenario mode. It measures multi-year persistence; newly planted cover has no history and scores ≈ 0.27 against a mature canopy's 0.95, so an honest live VSI would make *planting trees lower the score*. Freezing is disclosed in the UI.

The structural floor is a first-class output

EVI carries 0.35 of the weight and nothing the user does touches it. Under the additive form, Ballygunge's ceiling with a physically perfect retrofit — full canopy, 0.60 albedo, refuge adjacent, surface temperature at the rural baseline — was **61.4**; it could never leave "Moderate Resilience". The geometric form will differ and must be recomputed, but the property holds.

This is the product's sharpest output and must be **shown, not hidden**:

28 of Ballygunge's missing points are demographic exposure. Trees cannot fix them — that needs housing, health outreach and cooling-centre policy.

The UI displays achievable headroom separately from the structural floor. `structuralFloor(ward)` is a named export, not an incidental number.

6 · Uncertainty

The thermal model's measured error is ± 3.5 °C night / ± 5.0 °C day (`green-score-methodology.md` §5A). That propagates into any DC-URS value computed from modelled rather than observed temperature.

Under the additive form, 1 K of daytime LST moved the score ~ 1.0 point, so ± 5 K $\approx \pm 5$ points — a quarter of a tier band. **The geometric form changes this and it must be re-measured, not assumed.**

- **Baseline** scores use *observed* ECOSTRESS temperature → thermal uncertainty is the sensor's, not the model's.
- **Scenario** scores use *modelled* temperature → carry the full model band.

This asymmetry is deliberate and must be visible: the baseline is measured, the scenario is modelled.

`scripts/measure-dcurs-uncertainty.py` produces the band by Monte Carlo over the measured error distribution.

7 · Validation

Rank correlation against published Kolkata vulnerability maps is impossible at $n = 3$. That is a direct consequence of decision 1 and is stated rather than worked around. It becomes available when the ward set widens. Until then, four checks:

1. **Golden-case regression.** The source document's worked wards produce **Ballygunge 20.01** and **Baruipur 65.23** under the original engine. Recomputed under the corrected engine on the same inputs (2026-07-27, dry-run):

Ward	Original	Corrected	Tier
Ballygunge	20.01	21.13	Critical Hotspot
Barrackpore*	—	40.14	Vulnerable
Baruipur	65.23	64.16	Moderate Resilience

* Barrackpore inputs are plausible estimates, not measured — it has no worked example in the source document. It is included only to check tier separation and is **not** part of the frozen fixture until Phase 1 supplies real inputs.

The corrections cost almost nothing in calibration and buy a great deal in correctness. Spread falls only 45.2 → 43.0 points, the three wards still land in three distinct tiers, and the CEO's two anchor wards move by 1.1 and 1.1 points. Separately, the VSI guard turns water from 1.00 (perfect vegetation stability) to 0.00.

Freeze the two real wards as the fixture. Any later change that moves them must justify itself — the same discipline as `validate-model.mjs`. 2. **Face validity.** Ranking must match known Kolkata geography. A dense hot core scoring above a green fringe is a failure regardless of what the maths says. 3. **Sensitivity analysis.** Vary each of the nine indicators across its plausible range and record the swing in DC-URS. If one dominates, the other eight are decoration and the weighting is wrong. Report as a ranked table. 4. **Uncertainty band**, per §6.

Acceptance: all four run and reported. No numeric bar is set for face validity or sensitivity because none can be justified at $n=3$ — inventing one would repeat the ± 2 K mistake from the thermal calibration, where a bar was set before anyone knew what was achievable.

8 • File plan

src/data/		
wards.ts	NEW	ward list as data – widening is a config change
src/scripts/climate-engine/		
dc-urs.ts	NEW	the engine: pure, DOM-free, self-checked
dc-urs-inputs.ts	NEW	typed indicator record + provenance per field
scripts/		
fetch-worldpop.py	NEW	population density per ward footprint
fetch-census-2011.py	NEW	HVI_socio, areal-interpolated
fetch-sentinel-composites.py	NEW	seasonal FVC, albedo, VSI series via EE
compute-far.py	NEW	FAR from Google 2.5D + MS footprints
compute-tra.py	NEW	distance-to-refuge from OSM
build-dcurs-inputs.py	NEW	assemble → data/dc-urs/inputs.json
measure-dcurs-uncertainty.py	NEW	Monte Carlo band
validate-dcurs.mjs	NEW	the four §7 checks
data/dc-urs/		
inputs.json	NEW	one record per ward, every field with provenance
golden-cases.json	NEW	frozen regression fixture

`dc-urs.ts` carries `assertDcUrsLogic()` — bounded 0–100, geometric guard against zero pillars, VSI returns 0 for water, monotonicity in each pillar, and the frozen golden cases.

9 · Phasing

Phase	Work	Done when
0	<code>wards.ts</code> , <code>dc-urs.ts</code> with corrected maths, self-checks, golden cases recomputed and frozen	engine runs on hand-fed inputs, <code>npm run verify</code> green
1	Observed pillars: FVC, CanopyFrac, VSI, CRI, FAR, TRA	real ACI and partial EVI for three wards
2	<code>p_pop</code> from WorldPop, <code>HVI_socio</code> from Census 2011	EVI complete, vintages displayed
3	Validation — all four §7 checks	report published, sensitivity table recorded
4	Scenario layer + structural floor in the UI; Green Score retired	sliders move DC-URS; floor visible

Phase 0 is independent of all data acquisition and can start immediately.

10 · What could go wrong

Symptom	Response
Geometric form compresses all three wards together	Tested 2026-07-27: it does not. Spread 45.2 → 43.0 points, three distinct tiers retained. Re-check once real inputs land
Sensitivity shows one indicator dominating	Weighting problem, not a code problem. Report to CEO with the table; do not silently retune
Census 2011 units don't align with the 1400 m footprint	Areal interpolation with the assumption stated; if a ward's population is implausible, say so rather than smoothing it
WorldPop and Census 2011 disagree on population	They will — different vintages and methods. Use WorldPop, record the gap
Score moves a lot vs the Green Score	Expected and pre-agreed. Document the delta per ward
A pillar goes to exactly zero	The <code>[0.001, 1.0]</code> clamp catches it; log it, because it usually means missing data rather than a real zero

Sequencing rule

Same as the thermal calibration: **one change at a time, measured after each**. Every phase ends with a number that either matches expectation or does not. The failure this project has already seen twice — a constant tuned until the picture looked right — is avoided by the golden-case fixture and the sensitivity table, not by good intentions.